

MIRAGE: Optimal, Road-Graph-Native Location Obfuscation at City Scale for IoT Smart Cities

Pushkal Gupta, Praagya Garg, M. Nagasai Dattu, and S. Ebenezer Juliet

Abstract—IoT smart cities depend on continuous location reporting, and a large family of obfuscation mechanisms— k -anonymity, differential privacy (DP), graph constraints, temporal cloaking—has been proposed to protect it. Two questions remain unanswered in a deployable, quantitative way. First, how much privacy does each mechanism actually provide, on a common, comparable scale and against the threats it faces? Second, how far are these heuristics from the best achievable privacy on a real road network? We answer both. (1) We adopt an adversary-grounded privacy metric—the expected error of an optimal Bayesian adversary, in metres—and instantiate it under two threat models, a snapshot adversary and a Viterbi trajectory adversary with a Markov mobility prior, showing that a single privacy scale is inadequate because the two adversaries rank mechanisms differently. (2) We introduce MIRAGE, the first road-graph-native optimal obfuscation mechanism that scales to a real city. MIRAGE computes, per location, the release distribution that provably maximises the Bayesian adversary’s error subject to a utility budget, with the distortion and adversary error measured as graph shortest-path distance; it is made tractable by solving one small linear program per density-adaptive local region, turning an intractable $|V|^2$ program into $|V|$ tiny ones. On real OpenStreetMap networks of two cities under three mobility datasets (Beijing: GeoLife, T-Drive; Porto) we measure the price of heuristics: at matched utility, DP and k -anonymity leave 5–22% of the achievable privacy on the table in the operationally relevant regime, while MIRAGE is Pareto-optimal (privacy $\approx$ utility) up to the network’s intrinsic uncertainty ceiling. We further show rankings hold from a grid abstraction to the real network and that service availability is dataset-density-driven rather than mechanism-intrinsic. The complete framework—mechanisms, adversaries, datasets, and analysis—is released for reproducibility.

Index Terms—Location privacy, optimal obfuscation, Bayesian adversary, differential privacy, k -anonymity, road networks, trajectory privacy, Internet of Things.



1 Introduction

Smart cities collect fine-grained locations from IoT devices for analytics, traffic management, and emergency response, and simultaneously expose homes, workplaces, and mobility patterns [1]. Two decades of research have produced diverse obfuscation mechanisms: spatial k -anonymity [2], differential privacy and geo-indistinguishability [3], [8], graph-constrained perturbation [7], density adaptation [12], and temporal cloaking [2]. Yet a practitioner choosing a mechanism for a real deployment faces two unanswered questions.

Q1: How private is each mechanism, comparably and against the right threat? Mechanisms are scored on incompatible, family-specific scales ($k/k_{\max}$, a log-scaled ϵ , region areas) that have no common operational meaning and are usually measured against a single snapshot attacker, even for mechanisms designed to defeat trajectory tracking.

Q2: How far are deployed heuristics from the best achievable privacy? Classical theory shows that, for a given utility budget, there exists an optimal obfuscation mechanism—the solution of a linear program (LP)—that maximises a Bayesian adversary’s error [5]–[7]. But that theory has only ever been instantiated on the Euclidean plane or small grids, is snapshot-focused, and does not scale ($|V|^2$ variables), so it has never been evaluated on a real road network, and the

gap between deployable heuristics and the optimum on real topology is unknown.

We answer both questions.

Contributions.

- 1) An adversary-grounded, threat-model-aware privacy metric (Section 4). Privacy is the expected error of an optimal Bayesian adversary (Shokri et al. [4]), in metres, evaluated under both a snapshot adversary and a Viterbi trajectory adversary with a Markov mobility prior. We show the two adversaries induce different rankings, so a single privacy scale is invalid.
- 2) MIRAGE, the first road-graph-native optimal obfuscation mechanism that scales to a real city (Section 5). MIRAGE solves the optimal-mechanism LP with graph shortest-path distortion and on-graph support, made tractable by density-adaptive local LPs. DP and k -anonymity arise as constrained/degenerate special cases; an optional geo-indistinguishability constraint recovers a formal DP guarantee.
- 3) The price of heuristics on real road networks (Section 7). Across three mobility datasets on real networks of two cities we quantify, at matched utility, how much privacy DP and k -anonymity sacrifice (5–22% in the practical regime), with 95% bootstrap CIs and an ablation of MIRAGE’s knobs.
- 4) A reproducible, real-network, multi-city evaluation and open framework spanning a grid abstraction and real OSM graphs of Beijing and Porto, three

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datasets (GeoLife, T-Drive, Porto taxi), and a radio-technology sensitivity analysis of the energy model.¹

Honest positioning: we do not invent optimal obfuscation—that is due to Shokri et al. [5], [6] and Bordenabe et al. [7]. Our contribution is to make it graph-native, city-scale, and measured on real road networks, and to place it in a unified two-threat evaluation.

2 Related Work

Heuristic mechanisms. Spatial cloaking and k -anonymity [1], [2], [10], [11], geo-indistinguishability and the planar Laplace mechanism [8], [9], graph projection [7], density-aware k [12], and temporal cloaking [2] are all heuristics: they fix a rule and accept whatever privacy it yields.

Optimal mechanisms. Shokri et al. [5] showed the utility-optimal mechanism against a Bayesian localization adversary is an LP; [6] casts it as a Stackelberg game; Bordenabe et al. [7] add geo-indistinguishability as linear constraints. All are on the Euclidean plane / regular grids, are snapshot-focused, and are explicitly scalability-limited (the LP has $|V|^2$ variables); none is evaluated on a real road network. MIRAGE closes exactly this gap.

Trajectory correlation. Xiao and Xiong [13] showed per-report DP degrades under Markov temporal correlation; subsequent work handles trajectories via DP composition, dummies, or synthesis [14]–[16]. We use a Viterbi trajectory adversary to measure this degradation across mechanisms on a common scale.

Quantifying privacy. Shokri et al. [4] define location privacy as the expected error of an optimal adversary—the metric we adopt for a comparable, in-metres evaluation.

3 System Model and Threat Models

Graph. The environment is an undirected weighted graph $G = (V, E)$ with edge weights the road distance and d_G the shortest-path metric. We use two graphs over central Beijing: a controlled 30×30 grid (900 nodes) that removes topological confounds, and the real OpenStreetMap drive network for the same region, consolidated to 3,154 nodes / 5,111 edges with irregular degree (mean 3.24, max 23), 16% dead-ends, and hierarchy—absent from the grid.

Data. Three real mobility datasets across two cities. In Beijing: GeoLife [17], [18] (182 users, multi-modal GPS; ~ 11.4 M points map-matched to the real graph) and T-Drive (~ 10 k taxis; ~ 4 M points), a much denser regime. In Porto, Portugal: the ECML/PKDD taxi dataset (1.71 M trips, ~ 81 M points map-matched to a real Porto OSM graph of 3,455 nodes). Using multiple datasets of very different density, and a second city, separates mechanism properties from single-dataset or single-city quirks.

Two adversaries. Both know the mechanism and public background knowledge—a population prior $\pi(v)$ and a first-order Markov mobility model $P(v_{t+1} | v_t)$, estimated once from the data and shared, so no mechanism is advantaged. A snapshot adversary observes one release; a trajectory adversary observes a sequence and decodes it (Section 4).

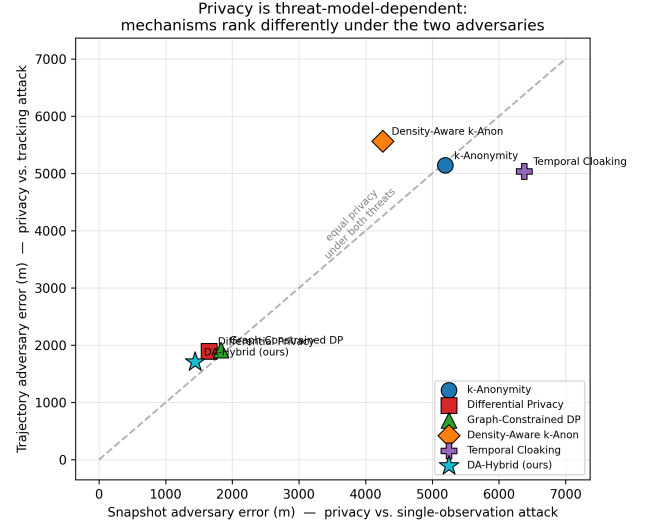


Fig. 1. Privacy is threat-model dependent: mechanisms rank differently under the snapshot (x) and trajectory (y) adversaries; points off the diagonal are protected asymmetrically.

4 Adversary-Grounded Privacy Metric

Privacy is the expected error of an optimal adversary that inverts the mechanism using public knowledge [4], measured in metres and thus comparable across all families.

Snapshot adversary. Observing release o , it forms the posterior $P(v | o) \propto P(o | v) \pi(v)$ and outputs the Bayes estimate $\hat{v} = \arg \min_{v'} \sum_v P(v | o) d_G(v', v)$; privacy is $\mathbb{E}[d_G(v_{\text{true}}, \hat{v})]$. The observation model $P(o | v)$ differs by mechanism, but the measured quantity is identical.

Trajectory adversary. Observing a sequence, it Viterbi-decodes the hidden node sequence, combining the per-report emission with the Markov prior; privacy is the mean tracking error and per-step re-identification rate. This is the threat temporal mechanisms target and that per-report noise fails against [13].

A single scale is inadequate. Fig. 1 plots the two errors against each other on the grid: density-aware adaptation and temporal cloaking swap rank between the threats, so any single-scale privacy comparison mis-ranks mechanisms defending different threats.

5 MIRAGE: Optimal Graph-Native Obfuscation

5.1 The Optimal Mechanism on a Graph

For a prior π and a utility (distortion) budget $D_{\max}$, the release distribution $f(o | v)$ maximising the optimal Bayesian adversary’s error, with both distortion and adversary error measured in graph distance d_G , is the LP

$$\begin{aligned}
 & \max_{f, y_o} \sum_o y_o && \text{(privacy, metres)} \\
 & \text{s.t. } y_o \leq \sum_v \pi(v) f(o | v) d_G(h, v) \quad \forall o, h && \text{(adversary)} \\
 & \quad \sum_v \pi(v) \sum_o f(o | v) d_G(v, o) \leq D_{\max} && \text{(utility)} \\
 & \quad \sum_o f(o | v) = 1, f \geq 0.
 \end{aligned} \tag{1}$$

The auxiliary y_o linearises the adversary’s inner minimisation (Shokri et al. [5]); we instantiate it with d_G and

1. <https://github.com/Pushkal-Gupta/Graph-based-Location-Privacy-IoT>

Algorithm 1 MIRAGE precompute (per graph, offline)

Require: graph G , prior π , budget $D_{\max}$, region size C

- 1: partition V into local regions (greedy, densest-first)
- 2: for each region R do
- 3: $d_R \leftarrow$ graph distances on R ; $\pi_R \leftarrow$ prior on R (renorm.)
- 4: $f_R \leftarrow$ solve LP (1)($d_R, \pi_R, D_{\max}$)
- 5: store $f_R(\cdot | v)$ for each $v \in R$
- 6: end for

on-graph output support, which is new. An optional geo-indistinguishability constraint $f(o | v) \leq e^{\varepsilon d_G(v, v')} f(o | v')$ adds a formal ε -DP guarantee, recovering [7] as a special case.

5.2 Scaling to a City: Density-Adaptive Local LPs

Program (1) has $|V|^2$ variables— $\sim 10^7$ for our graph—and is intractable directly. MIRAGE partitions V into density-adaptive local regions: starting from the highest-prior unassigned node, each region takes its C nearest still-unassigned nodes (dense areas are seeded first and covered by tighter regions). We solve one LP of size C per region and assign every member its optimal row. This turns an $|V|^2$ program into $\lceil |V|/C \rceil$ programs of $O(C^3)$ each—seconds to precompute offline, then $O(1)$ sampling at release time. Releasing across regions is never optimal under a distortion budget (it costs too much), so the partition is a principled, near-lossless approximation.

DP, k -anonymity, and our earlier density-adaptive hybrid (DA-Hybrid) are heuristic baselines; MIRAGE dominates them by construction and, we show, in measurement.

6 Experimental Setup

All mechanisms share the graph, data, prior, and distance layer. We evaluate two ways. (i) The frontier: for each mechanism we compute privacy (optimal- adversary error) and utility (expected distortion) analytically on the same regions, sweeping the mechanism’s knob ($D_{\max}$ for MIRAGE, ε for DP, k for k -anonymity), with 95% weighted-bootstrap CIs over regions. (ii) A deployment comparison: all seven mechanisms at representative operating points under the sampled snapshot and trajectory adversaries, reporting availability and spatial error. The energy model is swept across radio technologies (Section 7.4). MIRAGE uses $C=28$ unless noted.

7 Results

7.1 The Price of Heuristics

Fig. 3 shows the privacy-utility frontier on the real GeoLife graph. MIRAGE hugs the ideal (privacy=utility) at low distortion and dominates DP and k -anonymity throughout the practical regime; all mechanisms converge at high distortion to the network’s intrinsic prior-uncertainty ceiling (~ 720 m, the maximum error any mechanism can induce). Table 1 quantifies the gap: at matched utility, MIRAGE delivers +13 to +22% more privacy than the best heuristic in the 150–500 m regime on GeoLife, i.e., DP and k -anonymity waste that fraction of their distortion. The gap closes only past ~ 1100 m, where all saturate at the ceiling.

Cross-city generalisation. The same result holds on all three datasets and both cities (Fig. 2): MIRAGE’s gain

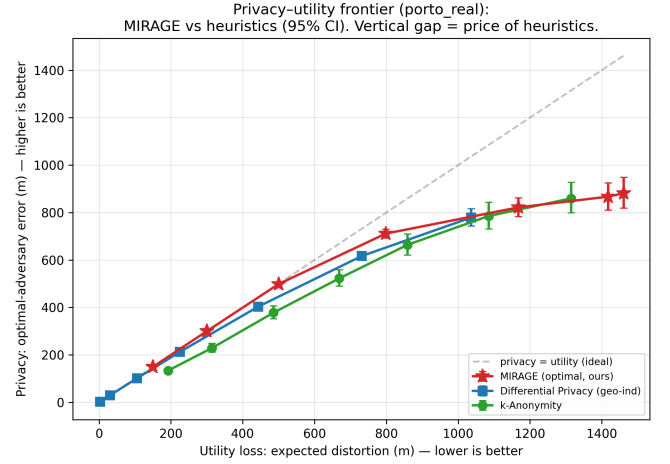


Fig. 2. Cross-city check: privacy-utility frontier on the real Porto (Portugal) network under the ECML/PKDD taxi dataset. MIRAGE dominates the heuristics here too; the price of heuristics is +5–12% in the practical regime.

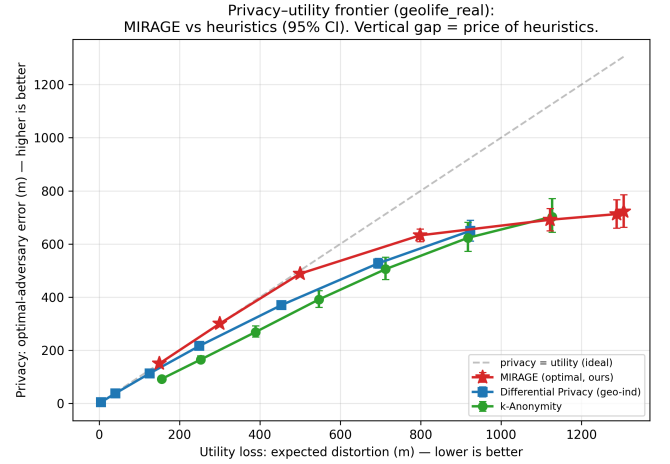


Fig. 3. Privacy-utility frontier (GeoLife, real graph, 95% CI). MIRAGE (optimal) dominates the heuristics; the vertical gap to MIRAGE is the price of heuristics. All converge at the intrinsic uncertainty ceiling.

over the best heuristic in the practical regime is +13–22% on GeoLife, +9–17% on T-Drive, and +5–12% on Porto. The magnitude tracks each city’s mobility structure—smaller where mobility is more predictable and heuristics are already closer to optimal—but MIRAGE is Pareto-optimal and dominant in every setting.

7.2 Ablation: What Makes MIRAGE Work

Table 2 ablates the two knobs. Region size C trades privacy for LP time: privacy rises with C (at $D_{\max}=800$: 421 m at $C=12$, 632 m at $C=28$, 715 m at $C=40$) while solve time grows as $O(C^3)$ (1.3 s to 21 s over all regions); $C=28$ is the knee. The optional geo-indistinguishability constraint costs some optimality (632 $\rightarrow$ 456 m at $\varepsilon/m=0.01$) in exchange for a formal DP guarantee—MIRAGE subsumes DP as its constrained case.

TABLE 1

Price of heuristics (GeoLife, real graph): privacy (optimal-adversary error, m) at matched utility loss.

Utility (m)	MIRAGE	DP	k -anon	MIRAGE gain
150	150	133	92	+13%
300	300	254	201	+18%
500	488	400	355	+22%
800	632	583	555	+9%
≥ 1100	700+	649	700+	$\sim 0\%$ (saturated)

TABLE 2

MIRAGE ablation (GeoLife, real graph).

Region size C	12	20	28	40
Privacy @ $D_{\max}=800$ (m)	421	551	632	715
LP time, all regions (s)	1.3	3.1	7.0	20.8

7.3 Deployment Comparison (All Mechanisms)

Table 3 compares all seven mechanisms on the real GeoLife graph under the sampled snapshot and trajectory adversaries. MIRAGE serves every user (100% availability), is on-graph, and provides strong balanced privacy; the two adversaries again reorder the heuristics. Two robustness findings emerge from repeating on T-Drive: rankings hold under the adversary metric, and availability is dataset-density-driven—on dense T-Drive every mechanism reaches 100% availability, whereas on sparse GeoLife k -anonymity (33%) and density-aware (61%) suffer denial of service (Fig. 4).

7.4 Energy-Model Sensitivity

The claim that radio transmission dominates per-report energy holds only for LPWAN/cellular radios (LoRa/NB-IoT/LTE-M, 99% radio share) but not for BLE, where computation is 6–50% of energy for BFS-based mechanisms (Fig. 5). MIRAGE’s release-time cost is a single categorical draw ($O(1)$), so it is as radio-dominated as DP.

8 Discussion

Guidance. MIRAGE gives, for a chosen utility budget on the operator’s own road network and mobility statistics, the release policy that provably maximises attacker error—up to +22% more privacy than tuned heuristics at equal utility, with 100% availability and on-graph outputs. Where a formal DP guarantee is required, its geo-indistinguishability-constrained variant applies at a small, measured cost.

Limitations (honest). (i) At high distortion the local-region cap (C) bounds MIRAGE’s reach, so heuristics catch up near the uncertainty ceiling; hierarchical or overlapping regions would extend the advantage. (ii) The optimal-mechanism LP is per-snapshot; the trajectory adversary shows temporal correlation still leaks, and a fully trajectory-optimal MIRAGE (optimising the sequential adversary via a mobility-corrected objective) is promising future work—the “sequential estimation” the name anticipates. (iii) Absolute error is graph-granularity dependent, so cross-topology conclusions rest on adversary-privacy and availability. (iv) The energy model is analytical, suited to ranking; we report its radio sensitivity explicitly.

TABLE 3

Deployment comparison (GeoLife, real graph, 10 min). Privacy = Bayesian-adversary error (m, higher = more private).

Mechanism	Snap. AE	Traj. AE	Avail.	Loc. Err
k -Anonymity	1694	3459	33%	4493
Differential Privacy	730	947	100%	466
Graph-Constr. DP	1339	732	100%	1400
Density-Aware	1437	3459	61%	2379
DA-Hybrid (heuristic)	311	1885	100%	563
Temporal Cloaking	2577	4636	100%	3215
MIRAGE (ours)	846	855	100%	1090

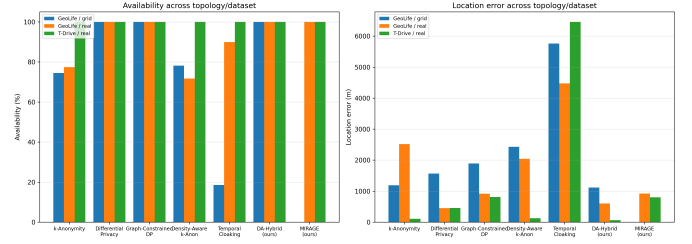


Fig. 4. Availability and error across the grid, the real graph (GeoLife), and the real graph under T-Drive. Availability rises to 100% for all mechanisms on the dense dataset—it is density-driven, not mechanism-intrinsic.

9 Conclusion

We measured location privacy on a common, adversary-grounded, threat-model-aware scale and brought optimal obfuscation to a real road network at city scale with MIRAGE. On real road networks of two cities across three datasets, deployed heuristics leave 5–22% of achievable privacy on the table in the practical regime, while MIRAGE is Pareto-optimal with full availability and on-graph outputs. Rankings hold from grid to real topology, and availability is shown to be dataset-density-driven. The framework is released for reproducible extension.

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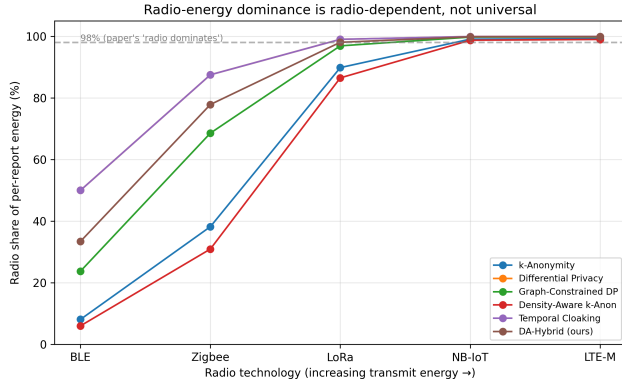


Fig. 5. Radio share of per-report energy vs radio technology. Radio dominance is radio-dependent, not universal.

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